EDVIRON DATA ANALYST ASSIGNMENT: EXPLANATION NOTE

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This document outlines the core logic, key business assumptions, and design rationale behind the EDVIRON Data Analyst solution. It provides a transparent audit trail for the financial model and interactive reporting components, ensuring accuracy, scalability, and ease of validation for stakeholders.

1. REVENUE LOGIC & DATA TRANSFORMATION

The primary technical challenge was the inconsistent pricing formats (Flat INR vs. Percentages). I built a robust transformation layer to ensure all financial metrics are absolute and audit-ready.

Standardization: Utilized Power Query to normalize disparate pricing types.

Formulaic Logic:

- **Flat:** Rate
- **Percentage:** (Transaction Amount * Rate) / 100

Revenue Streams:

- **ERP Revenue:** The margin between Merchant and Partner Pricing.
- **Edviron Net Revenue:** The margin between Partner Pricing and Edviron's Buying cost.
- **Edviron Gross Revenue:** The total platform "take" (ERP Revenue + Edviron Net).

2. CORE BUSINESS ASSUMPTIONS

To ensure the report's accuracy and adherence to financial standards, the following assumptions were applied to the data model:

Area	Assumption	Business Rationale
Revenue Recognition	Only "SUCCESS" status transactions are included.	Prevents artificial inflation of GMV by excluding failed or cancelled attempts.
Null Values	Missing pricing fields were treated as ₹0.00.	Maintains calculation integrity and assumes a zero-fee structure for those specific records.
User Metrics	Unique Users are defined by distinct Student Names.	Establishes a baseline for calculating payment frequency and user stickiness.
Risk / Exposure	Defined as "Success" transactions with "Capture Pending" status.	Identifies authorized funds that are not yet settled in the bank (Receivable Risk).

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3. DASHBOARD DESIGN RATIONALE

The dashboard follows a **Top-Down Visual Hierarchy** designed for streamlined executive decision-making:

- **KPI Tiles:** Strategically positioned at the top of the page to provide immediate visibility into core metrics: Total Volume, GMV, and Profitability (Net/Gross Revenue).
- **Dynamic Interactivity:** Essential filter Slicers (**Date, Partner, Gateway**) were integrated using Report Connections. This allows the end-user to filter the entire dashboard instantly, promoting interactive data exploration.

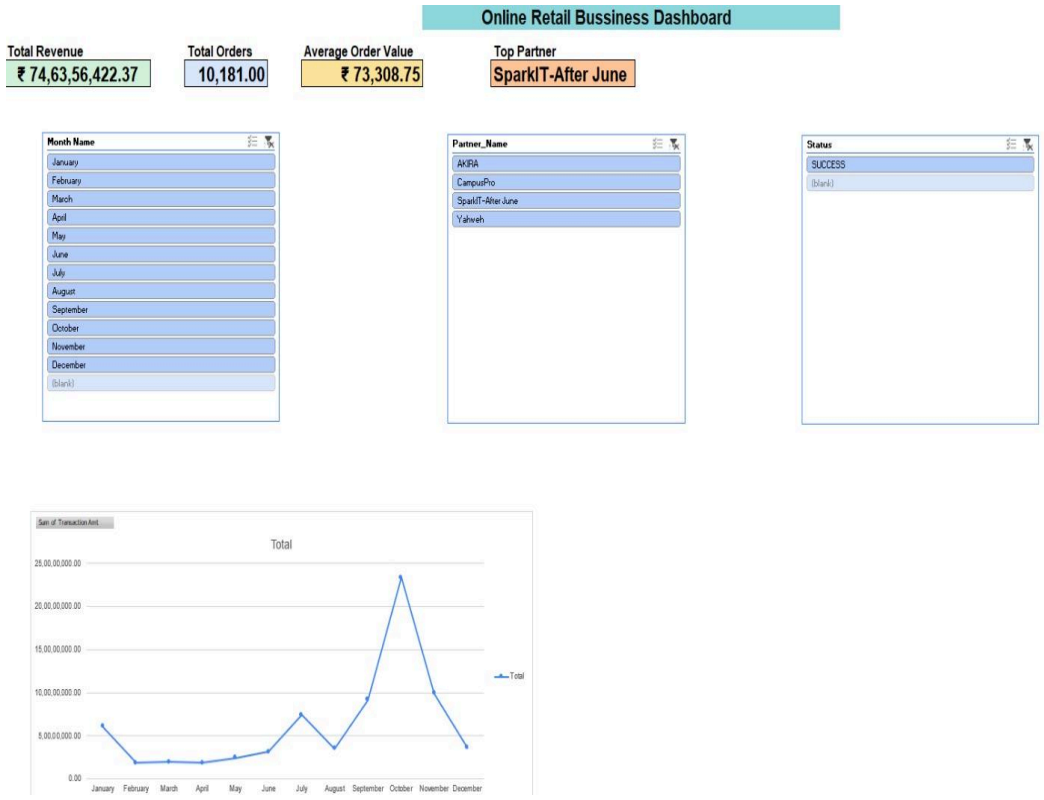
- **Trend Analysis:** A dedicated time-series chart identifies volume spikes and transaction patterns. This is critical for monitoring gateway stability and assessing peak load performance.

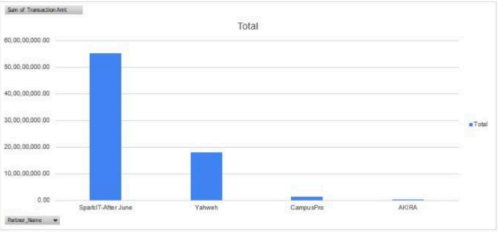
4. TECHNICAL IMPLEMENTATION (VBA & SCALABILITY)

The technical architecture of the workbook focuses on long-term sustainability and data integrity:

- **Modular Design:** The workbook is logically segmented into distinct layers: **Raw_Data** (original source), **Clean_Data** (transformed data), and **Calc_Model** (formula layer). This architecture maintains a "Single Source of Truth."
- **VBA Automation:** I developed custom macros to automate data ingestion and cleaning. This ensures the model is scalable and "future-proof," allowing for easy weekly data refreshes without manual intervention.
- **Dynamic Modeling:** No business logic is hard-coded. All revenue splits and pricing rules are formula-driven, allowing for rapid updates should partnership rates or gateway fees change in the future.

OUTPUT DASHBOARD:

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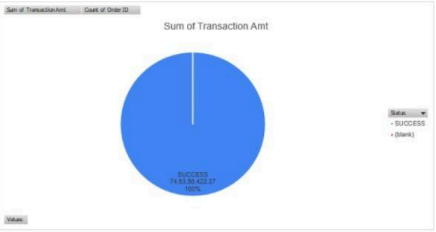
10,00,00,000.00
0.00
SparkIT-After June
Yahweh
CampusPro
AKIRA
Total
Partner_Name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Sum of Transaction Amt
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SUCCESS
(blank)
Values